

BIG VALLEY AGCM, AB, Canada

WMO: 710110

Lat: 51.9978N

Lon: 112.8025W

Elev: 870

StdP: 91.30

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-29.4	-26.5	-33.1	0.2	-29.1	-30.0	0.3	-26.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	28.7	16.5	26.7	15.6	24.9	15.2	18.3	25.1	17.2	24.1	16.3	23.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	12.6	21.4	14.6	11.5	20.5	13.5	10.7	19.3	55.5	25.0	51.9	24.4	48.9	23.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-32.8	32.1	2.6	2.5	-34.7	33.9	-36.2	35.3	-37.7	36.7	-39.6	38.5		
(5) N/A	N/A	N/A	-33.0	20.7	2.6	1.3	-34.9	21.7	-36.3	22.4	-37.8	23.2	-39.6	24.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-9.1	-10.0	-3.6	3.7	10.8	14.5	17.1	16.4	11.6	4.7	-3.7	-10.4
	DBStd	11.73	8.25	7.96	7.60	5.18	4.55	3.15	2.92	3.32	4.71	5.04	7.18	7.57
	HDD10.0	3075	593	561	422	194	45	3	0	1	35	175	413	633
	HDD18.3	5438	852	794	680	438	236	121	60	77	205	423	662	891
	CDD10.0	729	0	0	0	7	70	138	219	200	84	11	1	0
	CDD18.3	50	0	0	0	0	2	6	21	18	4	0	0	0
	CDH23.3	763	0	0	0	4	53	75	269	263	96	4	0	0
	CDH26.7	173	0	0	0	0	5	11	66	68	23	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	384	14	10	16	26	49	77	65	54	32	16	14	12
	PrecMax	490	34	24	54	60	93	152	151	108	82	35	48	29
	PrecMin	229	1	1	3	3	6	33	24	3	5	5	2	1
	PrecStd	65	8	6	10	14	24	30	30	28	19	9	11	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	7.5	14.2	22.1	27.2	28.0	31.0	31.2	29.6	22.7	15.0	7.4
		MCWB	3.0	3.1	6.3	9.5	12.8	15.4	17.9	16.8	14.6	11.3	7.7	3.0
	2%	DB	5.8	4.4	11.1	18.9	24.7	25.4	28.6	28.4	26.1	18.1	10.2	4.6
		MCWB	2.1	1.0	4.8	8.5	11.9	15.2	17.5	16.4	13.6	9.7	5.0	1.1
	5%	DB	3.8	2.6	8.4	15.6	22.2	23.5	26.2	26.3	23.5	15.4	7.2	2.2
		MCWB	0.8	-0.1	3.3	6.4	11.2	14.3	16.9	16.1	13.1	8.1	3.0	-0.7
	10%	DB	1.7	0.8	5.7	12.5	19.6	21.7	24.2	24.1	20.6	12.6	4.8	-0.1
		MCWB	-0.7	-1.3	1.9	5.1	10.0	13.4	16.5	15.7	11.9	6.5	1.4	-2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.1	7.3	10.4	13.8	18.1	20.6	19.3	15.9	12.1	8.0	3.4
		MCDB	7.1	6.9	13.5	21.6	23.9	24.5	27.1	25.5	26.2	21.2	14.3	6.5
	2%	WB	2.3	1.3	5.1	8.5	12.7	16.2	18.8	18.1	14.6	10.3	5.2	1.2
		MCDB	5.4	3.9	10.1	18.1	22.3	23.3	25.4	25.3	23.9	16.6	9.7	4.4
	5%	WB	0.9	0.1	3.5	7.0	11.8	15.1	17.6	17.1	13.4	8.6	3.4	-0.5
		MCDB	3.6	2.3	8.0	14.4	20.6	21.6	24.2	24.5	21.7	14.7	6.7	2.0
	10%	WB	-0.6	-1.2	1.9	5.5	10.7	14.1	16.7	16.0	12.4	7.0	1.7	-2.3
		MCDB	1.6	0.9	5.3	11.6	18.3	20.0	23.1	22.8	20.2	12.2	4.1	-0.3
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.5	9.7	11.2	12.0	11.2	12.0	12.5	12.5	11.2	9.2	8.4
		MCDBR	10.0	10.8	12.9	17.1	16.1	14.1	15.4	16.4	17.8	15.8	12.5	10.2
		MCWBR	7.3	8.1	7.7	8.5	6.9	6.3	6.8	6.8	7.8	8.5	8.0	7.7
	5% WB	MCDBR	9.9	10.7	12.5	16.0	14.3	12.5	13.4	14.0	16.0	15.1	11.9	10.1
		MCWBR	7.5	8.2	7.8	8.1	6.2	6.3	7.0	6.6	7.3	8.2	7.8	7.7
Clear-Sky Solar Irradiance	taub	0.265	0.269	0.285	0.330	0.350	0.354	0.352	0.366	0.315	0.281	0.253	0.243	
	taud	2.312	2.401	2.451	2.354	2.348	2.374	2.407	2.349	2.462	2.538	2.480	2.352	
	Ebn at Noon	751	870	923	908	901	898	894	859	874	848	783	725	
	Edh at Noon	61	75	88	110	116	115	109	110	87	65	51	49	
All-Sky Solar Radiation	RadAvg	1.09	2.13	3.52	4.73	5.75	6.11	6.46	5.25	3.67	2.20	1.15	0.83	
	RadStd	0.09	0.18	0.28	0.41	0.37	0.38	0.33	0.37	0.39	0.26	0.10	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page